Matthew Aragorn Pham

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EDUCATION

The University of Texas at Austin May 2027

Bachelor of Science in Computer Science, Minors in Statistics & Data Sciences and Business

Relevant Coursework: Operating Systems, Software Engineering, Artificial Intelligence, Machine Learning, Computer Architecture, Compilers, Data Structures, Algorithms, Discrete Math, Linear Algebra

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL, R, HTML, CSS
Tools: Docker, Kubernetes, Argo Workflows, Snowflake, AWS, ADLS, GitLab CI/CD, Git/GitHub, Linux, Postman
Libraries: React, Node.js, Scikit-learn, Pandas, NumPy, Bootstrap, REST APIs, Jupyter Notebook
Soft Skills: Problem-solving, Collaboration, Communication, Time Management, Adaptability

EXPERIENCE

William Blair – *Platform Engineer Intern* June 2026 – August 2026

- Architected automated Argo, Docker, and Kubernetes workflows to execute and validate large-scale data models.
- Engineered a scalable, cloud-native ETL data pipeline leveraging Python, Snowflake, and ADLS.
- Centralized disparate vendor workflows into a modular codebase using modern Git development workflows.
- Developed data validation algorithms for rigorous schema parity checks, ensuring dataset integrity.

Loan Ranger Capital – *Data Analyst Intern* May 2024 – July 2024

- Engineered Python and SQL data validation workflows, translating raw datasets into reliable, scalable metrics
- Integrated REST APIs for an automated system, enhancing infrastructure scalability and workflow efficiency.
- Built data collection and cleaning pipelines that expanded lead generation by 2,500+ contacts.

PROJECTS

Civic Housing – *Software Engineer* January 2026 – May 2026

- Engineered Node.js servers and RESTful APIs to serve structured data models from Postman-defined endpoints.
- Developed modular React components with Bootstrap to visualize real-time data and API endpoints.
- Designed relational database schemas with server-side pagination, multi-attribute filtering, and dynamic search.
- Deployed on AWS via GitLab CI/CD for automated verification and Postman for architecture documentation.

Pintos Operating System – *OS Engineer* August 2025 – December 2025

- Upgraded core OS Kernel components in C, debugging complex system-level memory and concurrency issues.
- Implemented robust multi-threading and synchronization primitives for secure CPU scheduling and memory safety.
- Architected virtual memory subsystems, including demand paging, frame allocation, and LRU eviction policies.
- Led a 3-person engineering team through weekly code reviews using Git source control and modern workflows.

High-Volume Predictive Modeling Pipeline – *Machine Learning Engineer* October 2025 – December 2025

- Engineered reusable Python data transformation modules to clean 250,000+ inputs and optimize execution latency.
- Applied feature engineering, cross-validation, and hyperparameter tuning to regression and ensemble models.
- Evaluated multiple ML pipelines using Scikit-learn, translating raw performance data into actionable insights.
- Automated data validation and model testing pipelines, improving reliability and reducing end-to-end run time.

System Emulator – *Systems Engineer* January 2025 – May 2025

- Built a five-stage pipelined CPU emulator in C across Fetch, Decode, Execute, Memory, and Writeback stages.
- Implemented data forwarding paths and pipeline stalling logic to resolve data, control, and structural hazards.
- Simulated hardware cache mechanisms and branch prediction to improve memory access latency and throughput.